Metaphorical Measure Expressions

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Outline

Canonical measure expressions

- In the concrete domain, measure expressions denote either a standard or a non-standard measure

(1) a. *metre*, *foot*, *litre*, *gallon*, etc.
    b. *glass*, *bucket*, *crate*, *truckload*, *pallet*, *mouthful*, etc.

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    - be pluralized
    - be freely combined with numeral expressions
    - govern plural verb agreement

(2) a. [Three crates of **sand**] were delivered to the store. **mass**
    b. [Three crates of **books**] were delivered to the store. **PL count**

Measuring in concrete and abstract domains

- Measure expressions can be: **SG indefinite** or **bare PL**

(3) Josie read {**a heap** | **heaps**} of **books** over the summer.

(4) Tony showered me with {**a heap** | **heaps**} of **assurances**.

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- **Nominal arguments** can denote concrete or abstract entities and be plural count or mass:

	PL count	mass
denote concrete entities	(3)	(5)
denote abstract entities	(4)	(6)

- In the abstract domain, there are no physical quantities to measure

Metaphorical measure expressions (MMEs)

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With a few exceptions (Klockmann 2017, 2020; de Vries & Tsoulas forthcoming), these expressions are seldom discussed in the literature

Metaphorical measure expressions (MMEs)

MMEs fall into two main semantic groups:

(7) a. **large quantity MMEs**: *bunch*, *heap*, *load*, *mass*, *oodles*, *scad*, *ton*, . . .

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- Speculatively: measure functions which do not denote what we intuitively and obviously view as large/small quantities cannot be used as measure functions of abstract entities

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(9) The FBI has heaps of information about Sam.

- They cannot be combined with numerals > 1 (unlike count Ns)

(10) a. The FBI and NSA each have a heap of info about Sam.
b. * The US gov't has two heaps of info about Sam.

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But neither of these involve counting *quantities*

- Instead, we're effectively counting *sources*,
- or just emphasizing (and not really counting at all)

Incomplete counting patterns: beyond English

This incomplete counting pattern shows up in other languages as well

- can be combined with the indefinite article
- can be pluralized
- cannot be combined with numerals > 1

(13) { ein Berg | (*drei) Bergen } von Sorgen (German)
‘{ a mountain | (*three) mountains } of worries’

(14) { un sacco | (*tre) sacchi } di odio (Italian)
‘{ a sack | (*three) sacks } of hatred’

(15) { kapka | (*tři) kapky } inspirace (Czech)
‘{ a drop | (*three) drops } of inspiration’

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  - Small quantity MME intensification is not available for MMEs derived from standard measures denoting a subjectively small quantity

(18) * Carly couldn't find an ounce of an ounce of info about camels.

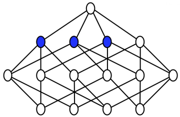
Outline

Quantization and Cumulativity

(Link 1983; Krifka 1986, 1989; Bach 1986)

Quantization: No member of the set is a proper-part of another

$$QUA(P) \leftrightarrow \forall x \forall y [P(x) \wedge P(y) \rightarrow \neg x \sqsubset y]$$



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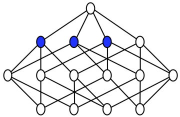

Cumulativity: The sum of any two members of the set is also a member of the set

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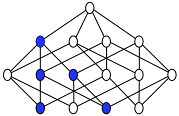

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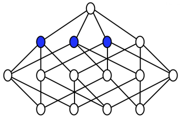

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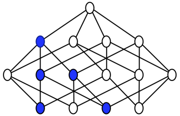

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Quantization and Cumulativity are inconsistent for non-singletons:

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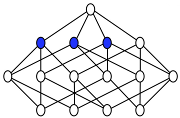

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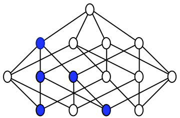

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But: $\neg QUA(P) \nvDash CUM(P)$ and $\neg CUM(P) \nvDash QUA(P)$

- So some sets are neither QUA nor CUM, e.g., $\{a, b, a \sqcup b \sqcup c\}$

The structure of the lexicon

(Sutton & Filip 2020; Landman 2011, 2016)

- The interpretations of NPs have a bipartite structure — two sets, each determined relative to the context of utterance
  - i.e. the interpretation of every NP has the form:

$$\langle \textbf{extension}, \textbf{counting\_base} \rangle$$

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- the extension — what the NP is standardly taken to apply to at world *w* and context *c*
- the counting base — a set of entities that are accessible to modification with numerals, determiners, distributive modifiers etc.
  - these entities are 'countable' if this set is quantized at world *w* and context *c*

The count/mass distinction

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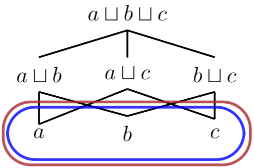
SG Count

e.g., *apple*, *book*



counting base

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extension

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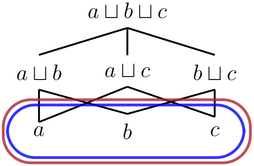

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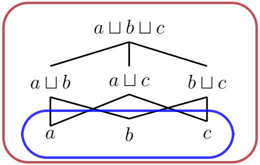

counting base
$\subset$
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counting base is quantized, extension is cumulative

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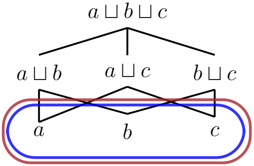

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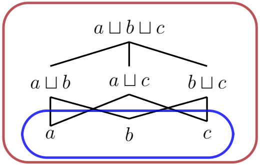

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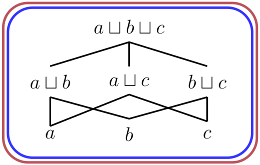
Mass

e.g., *air*, *mud*



counting base
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Both sets are cumulative and not quantized

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e.g., for $[\![\text{books}]\!]$ in (20): the extension set, ${}^*\mathcal{Q}_c(BOOK_w)$, is the cumulative set of single books and sums thereof. The counting base set, $\mathcal{Q}_c(BOOK_w)$, is the set of single books, a quantized set.

Accessing the extension and counting base sets

We use **ext** and **cbase** as the first and second projection functions of the tuples in such lexical entries. For example:

(22) $[\![\text{books}]\!](c)(w)(a) = \left\langle\, {}^{*}\mathcal{Q}_c(BOOK_w)(a),\ \ \lambda y.\, \mathcal{Q}_c(BOOK_w)(y) \,\right\rangle$

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$[\![\text{book(s)}]\!]$ is count, since for every context, $\textbf{cbase}([\![\text{books}]\!](c)(w)(x))$ is a quantized set

- i.e., $\lambda y.\mathcal{Q}_c(BOOK_w)(y)$ is a quantized set

The grammatical reflexes of countability

- In number-marking languages with indefinite articles at least, count nouns can be modified by numeral expressions,

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- Singular count nouns can be pluralised and combined with the indefinite article, while plural count nouns and mass nouns cannot:

(27) I bought a book / books.

(28) I bought # a books / # bookses.

(29) I felt # an air / # airs.

Quantization, cumulativity and the grammatical reflexes of countability

	Example	C. base set	Num Mod	Extension set	Indef	PL morph
SG count	*cat, bowl of apples*	QUA	Y	QUA, $\neg CUM$	Y	Y
PL count	*cats, bowls of apples*	QUA	Y	$\neg QUA$, CUM	N	N
mass	*mud*	$\neg QUA$	N	$\neg QUA$, CUM	N	N

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- Counting base set is quantized $\Rightarrow$ Modification with numerals > 1
- Indefinite article & PL morphology? We have two options:
  1. Extension is quantized
  2. Extension is not cumulative (a weaker condition)

[Recall: a set can be neither cumulative nor quantized]

More categories of countability?

The theory predicts a logical space for some category **X** / **PL X**:

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- The grammatical reflexes of this category, if attested, will decide between the two restrictions for PL morphology and the indefinite article (*QUA* or ¬*CUM* extension)

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- The grammatical reflexes of this category, if attested, will decide between the two restrictions for PL morphology and the indefinite article (*QUA* or ¬*CUM* extension)
- Spoiler: MMEs are examples of such a category

Outline

Overview of analysis

MMEs (e.g., *heap/iota (of)* used metaphorically) share some properties with the literal uses of *amount (of)* or *quantity (of)*. (See, e.g., Chierchia's (2010) analysis)

- they specify an amount of P relative to a context
- they display 'pseudo-cumulativity' — cumulative in some contexts, but unlike mass nouns, they are **not cumulative at every context**

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i. measure phrases with MMEs (*MME of P*) do not specify a disjoint or quantized set of P at all contexts, hence they are not generally compatible with numeral expressions in counting constructions
ii. small-quantity MMEs lexically encode a small quantity of P, and large quantity MMEs a large quantity of P, relative to the context

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MMPs fit our missing category!

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Quantity and *Amount*

Chierchia 2010: *amount (of)* and *quantity (of)* encode a contextual partition operator Π_c, such that for all P, $\Pi_c(P)$ is a disjoint (and so quantized) subset of P:

(30) $[\![\text{amount}]\!] = \lambda c \lambda P \lambda x.\Pi_c(P)(x)$

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Earlier precursors of the idea that these expressions are quantized only relative to a context of use: (Krifka 1998; Zucchi & White 1996, 2001). See also variants in Landman 2011

Recap: features of MMEs

- Unlike non-metaphorical uses of *quantity* and *amount* of P, *MME of P* does not specify a disjoint or quantized set of P at all contexts, and so is not generally compatible with numeral expressions in counting constructions

Recap: features of MMEs

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i. Unlike non-metaphorical uses of *quantity* and *amount* of P, *MME of P* does not specify a disjoint or quantized set of P at all contexts, and so is not generally compatible with numeral expressions in counting constructions

ii. Unlike non-metaphorical uses of *quantity* and *amount*, small quantity MMEs encode a small quantity of P, and large quantity MMEs a large quantity of P, relative to the context

- Accounting for ii. will already give us an explanation for i.
- But: We also need to explain pseudo-cumulativity: MMPs are not cumulative in every context
  - Prima facie tension with ii. for large quantity MMEs

Contextually large/small amounts

A measure function μ of polymorphic type $\langle n, \langle \alpha t, \alpha t \rangle \rangle$:

- e.g., $\langle n, \langle et, et \rangle \rangle$, $\langle n, \langle \langle st, t \rangle, \langle st, t \rangle \rangle \rangle$, etc.
- $\mu_{\mathbf{s}}(x, P) = n$ states that the magnitude of x on a scale $\mathbf{s}$ with respect to P is n

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Large quantity MMEs: the quantity of P is specified to be above some contextually specified P-threshold $n_{c,P}$

$$\mu_{\mathbf{s}}(x, P) > n_{c,P}$$

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$$\mu_{\mathbf{s}}(x, P) < n_{c,P}$$

Context-indexed covers

Non-cumulativity is ensured via a context-indexed operator Δ_c of type $\langle et, et \rangle$:

(31) $\forall c \forall P \exists X [\Delta_c(P) = X \wedge X \subseteq {}^*P \wedge \sqcup X = \sqcup P]$

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- A subset of ${}^{*}P$ with the same supremum as P
- $\Delta_c(P)$ is a set that is akin to a cover of P in the sense of Gillon 1987. Not only a minimal cover!

Example: $\Delta_c(\{a, b, a \sqcup b\}) \Rightarrow$ 5 possible context-indexed covers:

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Only 2 of these covers are quantized: $\{a, b\}$, $\{a \sqcup b\}$
And only 2 are cumulative: $\{a \sqcup b\}$, $\{a, b, a \sqcup b\}$

Lexical entries

(32) $[\![\text{heap}_{\text{MME}}]\!] =$

$$\lambda\mathcal{P}\lambda c\lambda w\lambda x.\left\langle \begin{array}{l} \Delta_c(\lambda y.\mathbf{ext}(\mathcal{P})(c)(y))(x) \wedge \mu(x, \mathbf{cbase}(\mathcal{P})(c)(x)) > n_{c,\mathcal{P}}, \\ \lambda y.\Delta_c(\mathbf{cbase}(\mathcal{P})(c)(x))(y) \wedge \mu(y, \mathbf{cbase}(\mathcal{P})(c)(x)) > n_{c,\mathcal{P}} \end{array} \right\rangle$$

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Example: the interpretation of *info* is a set of (sums of) propositions (Sutton & Filip 2020); the counting base set for *heap/iota of info* counts as a large/small amount of info in the context

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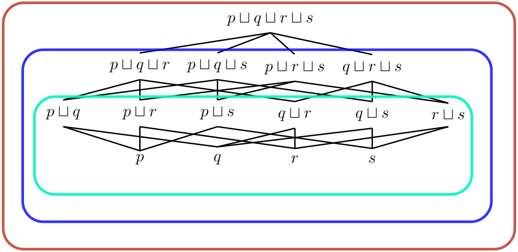
For some c':



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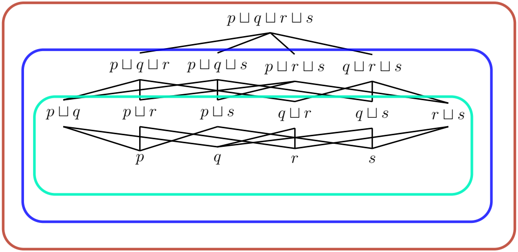

$\mathbf{INFO_w}$

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Counting base at $c' = \lambda x.\Delta_{c'}(INFO_w)(x) \wedge \mu(x, INFO_w) < n_{c',[\![\text{info}]\!]}$

- $\neg CUM$, $\neg QUA$
- In c', *iota of information* is not countable, but can be pluralised or used with the indefinite article

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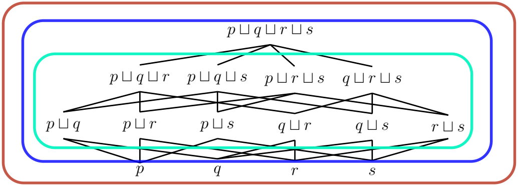

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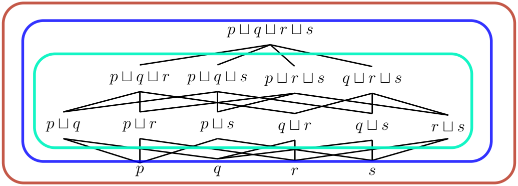
For some c':



$\mathbf{INFO_w}$
$\mathbf{\Delta_{c'}(INFO_w)}$
$\boldsymbol{\lambda x.\mu(x, INFO_w) > n_{c',[\![info]\!]}}$

Counting base at $c' = \lambda x.\Delta_{c'}(INFO_w)(x) \wedge \mu(x, INFO_w) > n_{c',[\![\text{info}]\!]}$

- $\neg CUM$, $\neg QUA$
- In c', *heap of information* is not countable, but can be pluralised or used with the indefinite article

Outline

Large quantity MMEs and pseudo-cumulativity

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(36) After experiments 1 and 2, Alex and Billie each had a heap of data / Alex and Billie together had heaps of data.

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In this sense large quantity MMEs are only *pseudo-cumulative*

Context-shifts in the concrete domain

(Rothstein 2010; Chierchia 2010; Sutton & Filip 2016)

Context shifts can be physical:

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But context-shifts can also be conceptual:

- We can discern either two heaps or one without affecting any change to the sand
- Counting perspectives aided by physical/perceptual properties of the entities, e.g., shape

Context-shifts in the abstract domain

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- For this reason, perhaps large quantity MMEs have a cumulative flavour because switching between contexts/perspectives is so easy
- Further work: to discern abstract entities, do we need a conceptual anchor in the concrete domain? (Grimm 2014; Sutton & Filip 2019)
    - e.g., discerning between heaps of info via agents or sources, etc.

Small quantity MMEs and plural morphology

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(39) . . . who are the Young Avengers anyway? What are their powers? We get only **slivers** of information.

(40) . . . the little **glimmers** of beauty, joy and love that make life sweet regardless of circumstances.

(41) In an entirely futile attempt to preserve the few remaining **iotas** of Danny’s good character. . .

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- Any awkwardness in some examples we attribute to a clash between the semantics of the small quantity MME and a pragmatic abundance inference sometimes triggered by plural morphology

Arguments against ambiguity
(de Vries & Tsoulas forthcoming)

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dV&T: MMEs do not simply involve a vague or figurative use of the measure expression, since this would not account for the countability differences between the literal and metaphorical use

- Response: We have shown how these measure expressions can be polysemous between literal and metaphorical senses, and account for their distinct distributional patterns

Consequences for countability theories

Further support the view that quantization and cumulativity relative to context are one of the keystones for modelling the wide variety of countability data

- Given that there is a logical space between non quantization and cumulativity (a predicate can be $\neg QUA$ and $\neg CUM$), a theory of countability based on QUA and CUM should predict that some expressions fill this logical space
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We need, minimally, two sets: extension and counting base

- In order to capture the selectional restrictions of both numeral expressions, on the one hand, and PL morphology and the indefinite article on the other, one set is not enough

Thanks and acknowledgements

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Non-number numerals

Our MMEs involve the recruitment of measure expressions in order to express non-measured 'amounts', just to express largeness/smallness

Something similar happens when

1. real numerals are used hyperbolically, or

   (42) I swear, there were a hundred dogs at the park!

2. 'fake' numerals are invented

   (43) There must have been a bajillion dogs there!

   - *zillion*, *kajillion*, ...
   - See Chrisomalis 2016

In either case, we just express largeness/smallness, not a specific amount

Challenges for a compositional analysis

Perhaps polysemy is not a satisfying enough explanation, and we should want a compositional derivation of the metaphorical interpretation from the literal one. We do not think so:

- Part of $[\![\text{heap (of)}_{\text{literal}}]\!]$ is plausibly a conjunction such as $\lambda P \lambda x. \ldots HEAP(x) \wedge P(x)$. Mutatis mutandis for $[\![\text{pile (of)}_{\text{literal}}]\!]$, $[\![\text{sliver (of)}_{\text{literal}}]\!]$ etc.
- But if *HEAP* is of type $\langle e, t \rangle$ and encodes e.g., shape and size information, it is not obvious what kind of function could map this literal meaning to the metaphorical one
  - I.e. no function that maps $\lambda P \lambda x. Q(x) \wedge P(x)$ to $\lambda P \lambda x. P(x) \wedge \mu(x, P) > n$
    (We can't simply delete the HEAP, PILE, SLIVER etc. from the literal meaning via a compositional process.)

Challenges for a compositional analysis cont.

- If, alternatively, *HEAP*, *SLIVER*, etc., are of polymorphic types that can apply to sand, metal and information, courage, etc., then both the metaphorical and literal senses could involve these predicates
- However, then we would have a hard time explaining differences in inference patterns between literal and metaphorical uses:

(44) *There are heaps of sand.* $\vDash$ *The sand is in heaps.*

(45) *There is a sliver of metal.* $\vDash$ *The metal is a sliver.*

(46) *We received heaps of information.* $\nvDash$ *The information is in heaps.*

(47) *We gained a sliver of knowledge.* $\nvDash$ *The knowledge we gained is a sliver.*

(48) *You need heaps of sand to make an artificial beach.* $\nvDash$ *The sand you need must be in heaps.*